

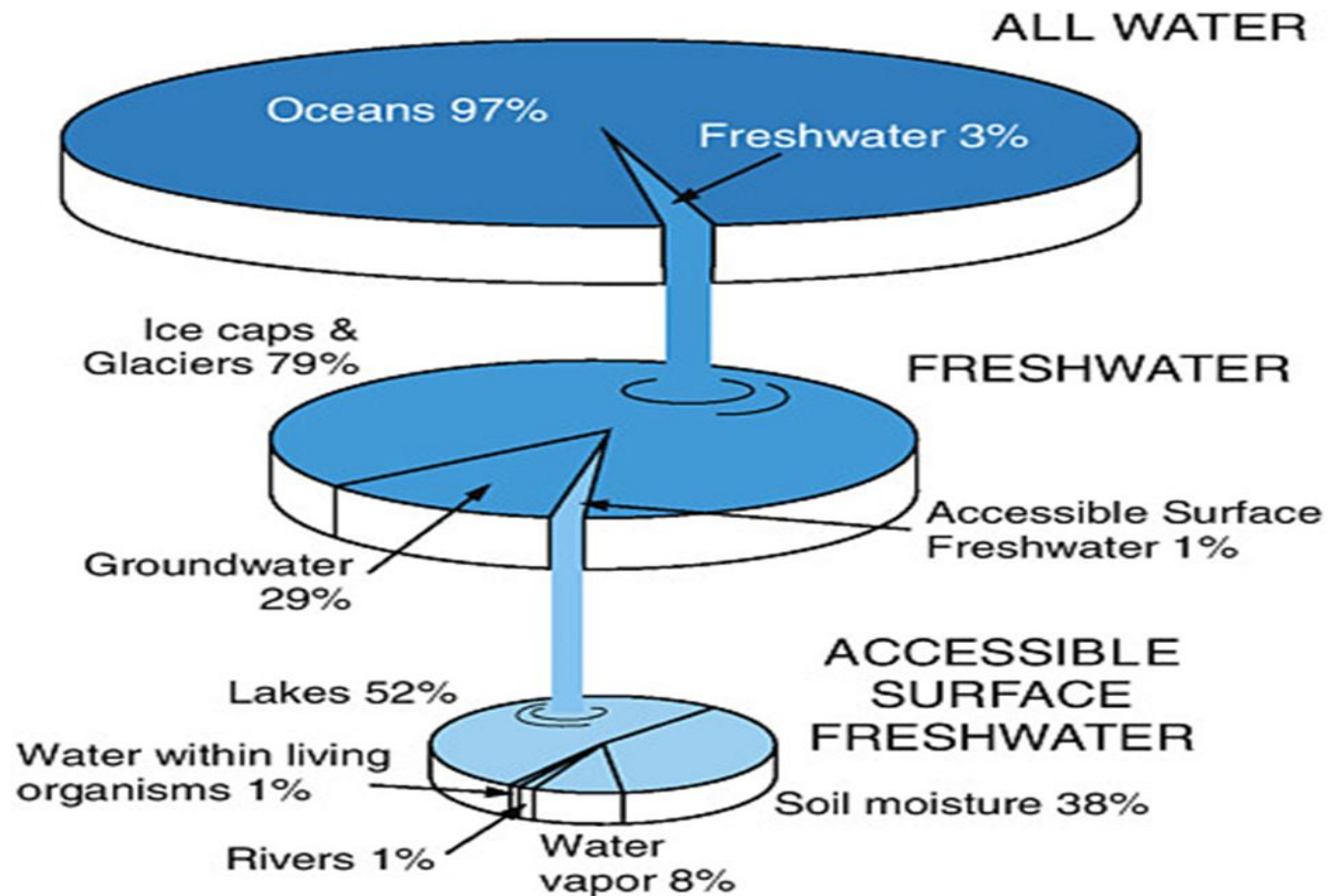
Water Resource





Water Resource

Distribution of Global Water





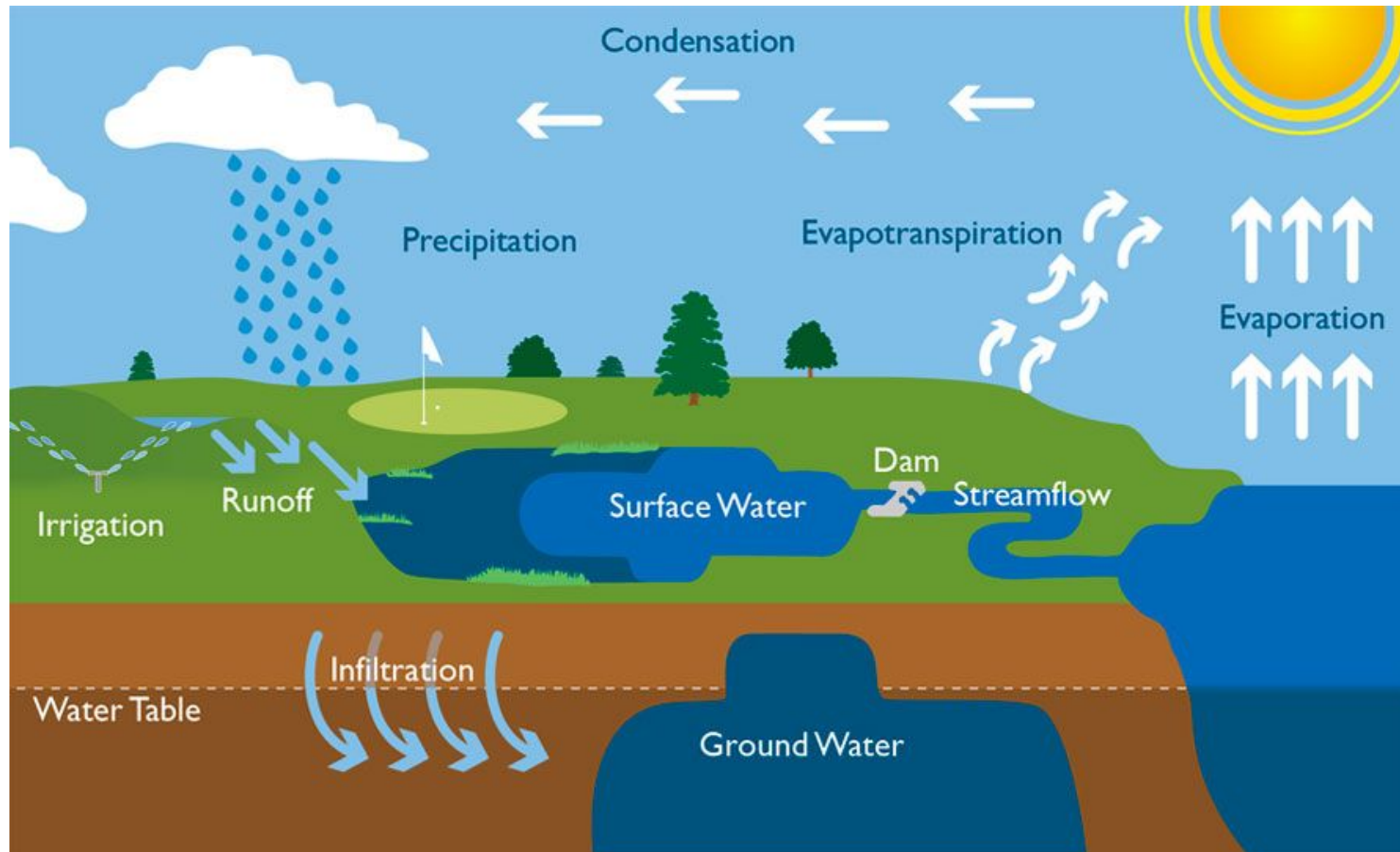
Water Resource

Hydrological Cycle!!!



Water Resource

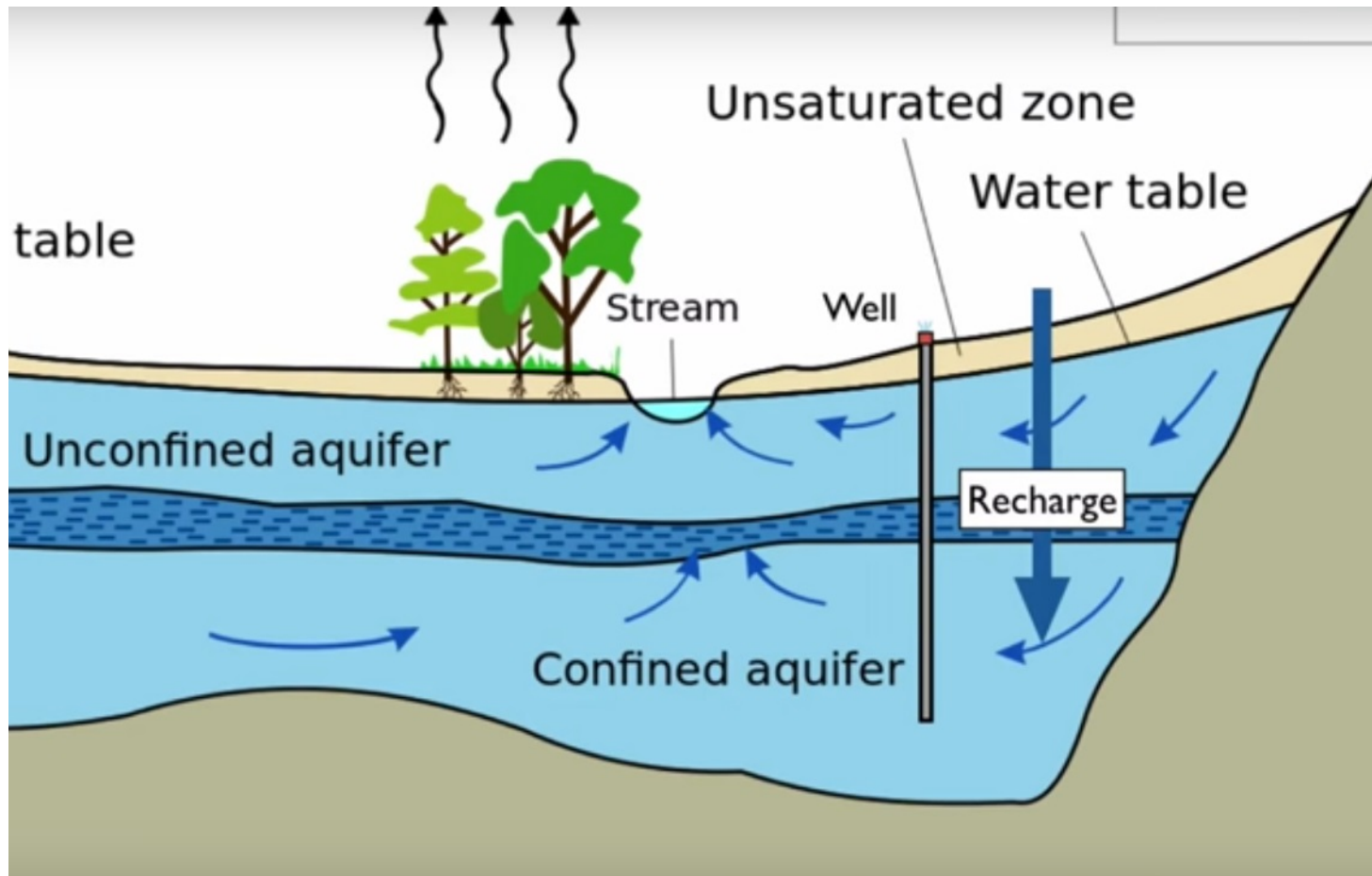
Hydrological Cycle





Water Resource

Fresh water





Water Resource

Water Use

Agriculture – 70%

Industry – 22%

Domestic – 8%

Water Storage Ground water – Aquifer

Helps!!

Potable water to drink

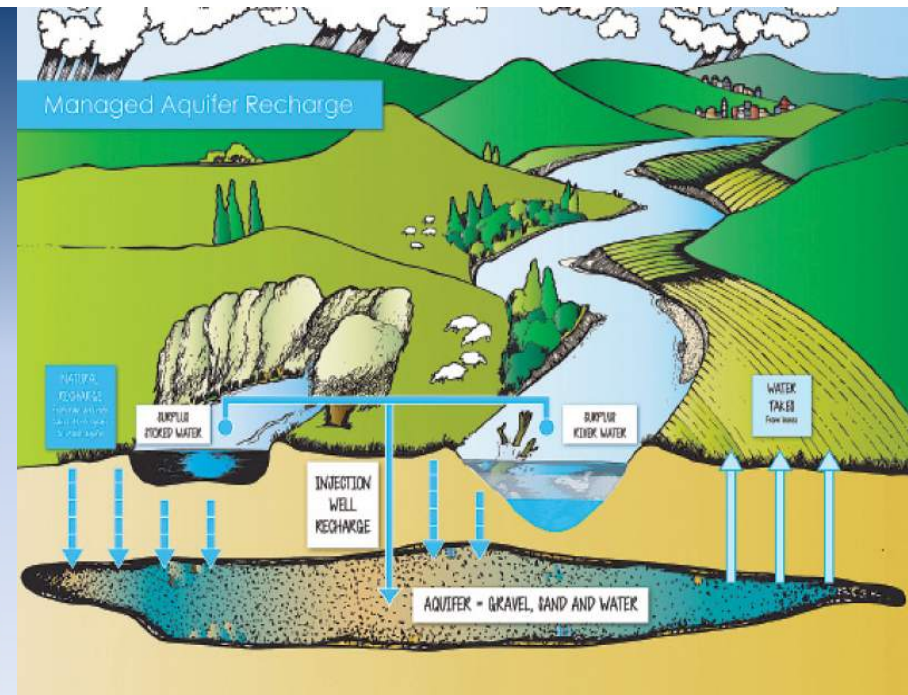
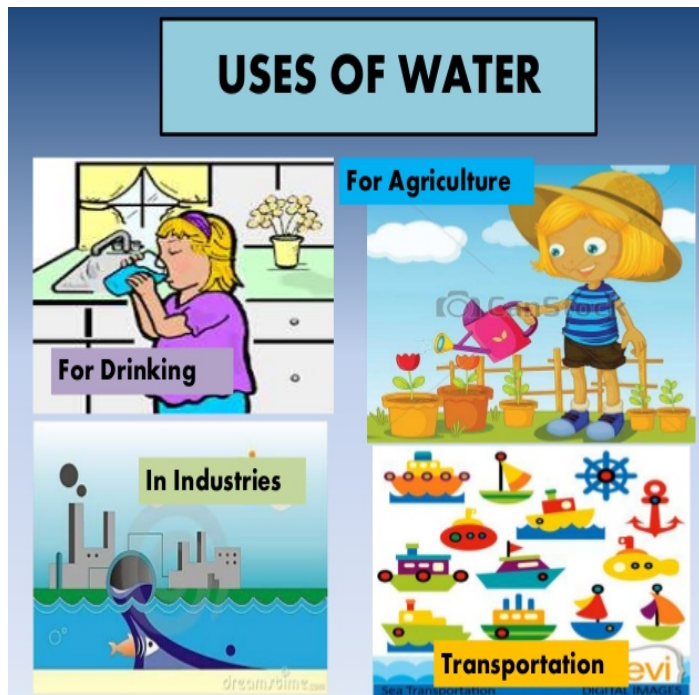
Reduce, Reuse, and Recycle

(3R)

Concern!!

Ground water depletion

Untreated ground water





Water Resource

Water Storage – Surface water - Reservoir

Helps!!

Energy Production

Reduce Flood

Irrigation

Reclaimed water

Concern!!

Destruction

Less Evaporation

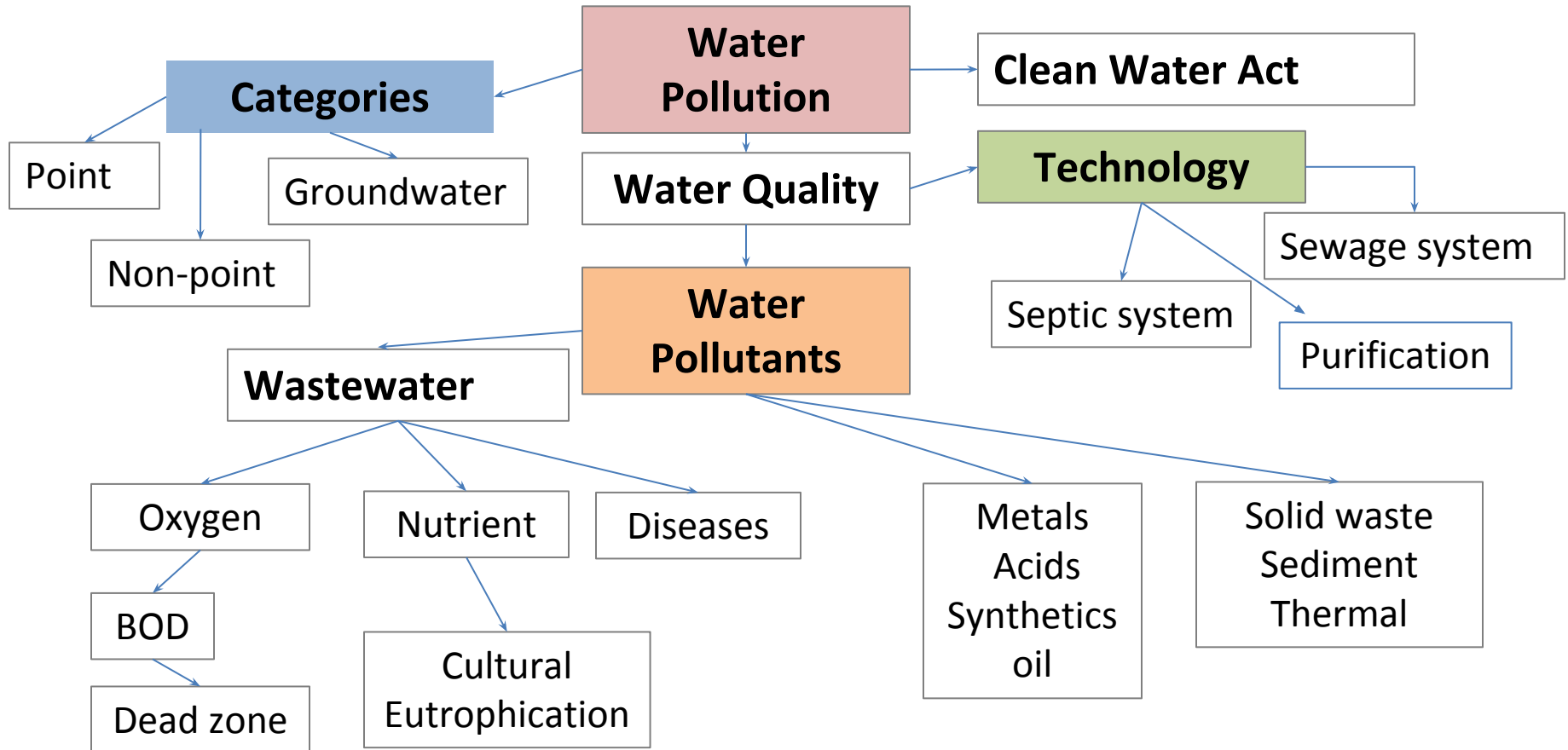
Wildlife hampered

Eviction of people or community





Water Resource





Water Resource

Water Pollution

The contamination of surface water (streams, rivers , lakes, oceans) or groundwater with substances produced through human activities and natural phenomenon that negatively affect organisms.

Pollution Categories:

Point sources – distinct locations that pump waste or contaminants into a waterway. You can target (or point to).

Non-point sources – diffuse areas such as an entire farming region that pollutes a waterway. Hard to target where the source exactly is.

Groundwater pollution/depletion - long-term water-level/quality declines caused by sustained groundwater pumping a/o contamination



Water Resource

Categories



Point



Non-point



Groundwater



Water Resource

Water pollution

Causes

Sewage

Nutrients

Waste water

Chemical waste

Radioactive waste

Oil pollution

Impacts

Death of aquatic (water) animals

Disruption of food-chains

Diseases

Destruction of ecosystems





Water Resource

Water quality

“Water quality” is a term used here to express the suitability of water to sustain various uses or processes. Any particular use will have certain requirements for the physical, chemical or biological characteristics of water (UNEP/WHO, 1996)

Water quality – parameters:

- **Physical parameters.** Colour, odour, turbidity, taste, temperature and electrical conductivity
- **Chemical parameters.** the amount of carbonates, sulphates, chlorides, fluorides, nitrates, and metal ions
- **Biological parameters.** Quantity Existence of algae, fungi, viruses, protozoa and bacteria



Water Resource

Water quality and dissolved oxygen (DO) content in parts per million (ppm) at 20°C

BOD Level in mg/liter	Water Quality
1 - 2	Very Good: There will not be much organic matter present in the water supply.
3 - 5	Fair: Moderately Clean
6 - 9	Poor: Somewhat Polluted - Usually indicates that organic matter present and microorganisms are decomposing that waste.
100 or more	Very Poor: Very Polluted - Contains organic matter.

- Only a few fish species can survive in the water less than 4ppm at 20°C



Water Resource

Water Pollutants

Types:

- (i) Sewage Pollutants (Domestic and Municipal Waste)
- (ii) Industrial Pollutants
- (iii) Agricultural Pollutants
- (iv) Radioactive and Thermal Pollutants

In the form of:

- ☐ Human Wastewater
- ☐ Chemical pollutants
- ☐ Non-chemical pollutants



Water Resource

Human Wastewater

Three reasons scientists are concerned about **waste water**

1. Oxygen demanding waste like bacteria (**E.coli**) that put a large demand for oxygen in the water
2. Nutrients that are released from wastewater decomposition can make the water more fertile causing eutrophication
3. Wastewater can carry a wide variety of disease-causing organism.

Eutrophication, defined as the addition of 'excess' nutrients to a water body, is a widespread environmental problem facing the world's aquatic habitats.



Water Resource

Biochemical Oxygen Demand (BOD)

BOD – the amount of oxygen a quantity of water uses over a period of time at a specific temperature

- Lower BOD values indicate the water is less polluted and higher BOD values indicate it is more polluted by wastewater
- When bodies of water have high oxygen demand due to microbial decomposition, the amount of O_2 for other organisms can be very low



Water Resource

Wastewater



Decomposers



Dead zone



Water Resource

Eutrophication

Eutrophication is an abundance of fertility to a body of water. For instance, Algae bloom

- caused by an increase in nutrients (including nitrogen and phosphorus), such as fertilizers.
- can cause a rapid growth of algae which eventually dies, causing the microbes to increase the BOD.



Water Resource

Cultural Eutrophication

Algae



Dead zone



Water Resource

diseases

Common diseases from Human Wastewater

- ✓ Cholera
- ✓ Typhoid fever
- ✓ Stomach flu
- ✓ Diarrhea
- ✓ Hepatitis

Facts!

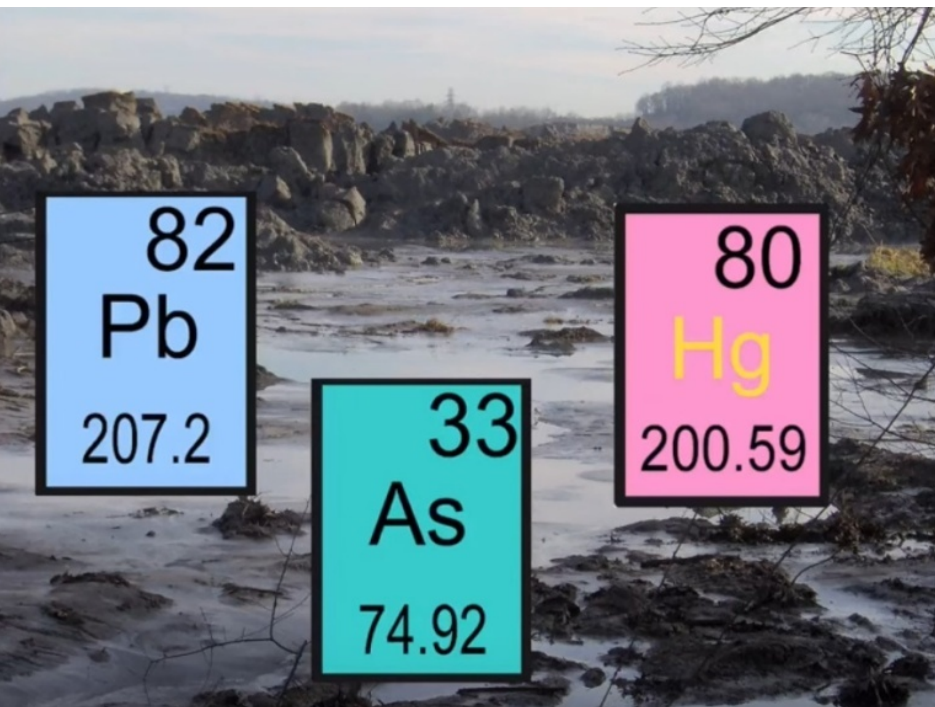
1/6 of the world's population does not have access to sufficient supplies of drinking water

1/2 of the diarrheal and malaria deaths could be prevented with safe drinking water, and proper sanitation and hygiene



Water Resource

Chemical Pollutants



Non-chemical Pollutants

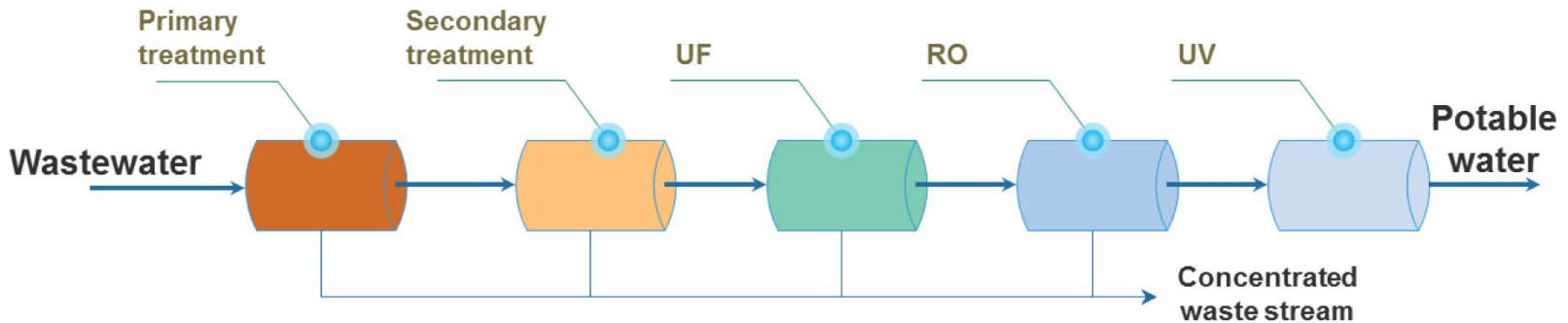




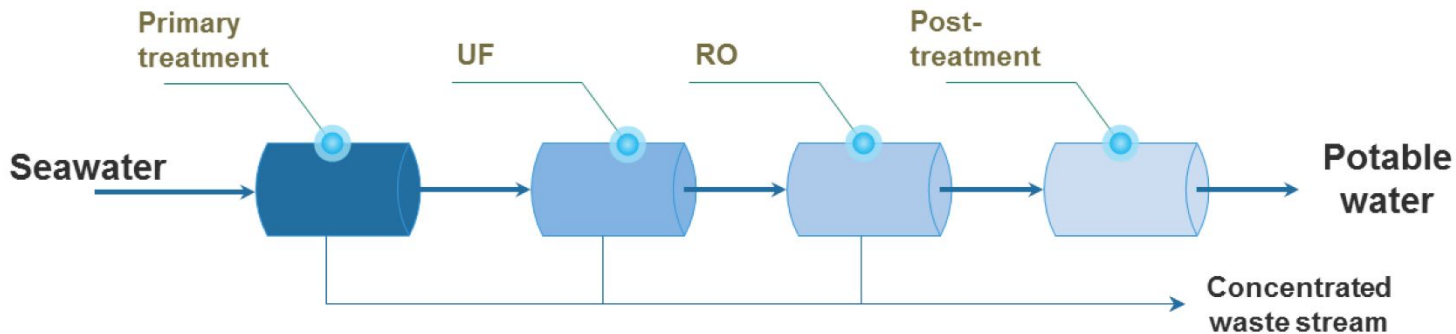
Water Resource

Technology Used to Water Treatment

Potable Water reuse treatment train



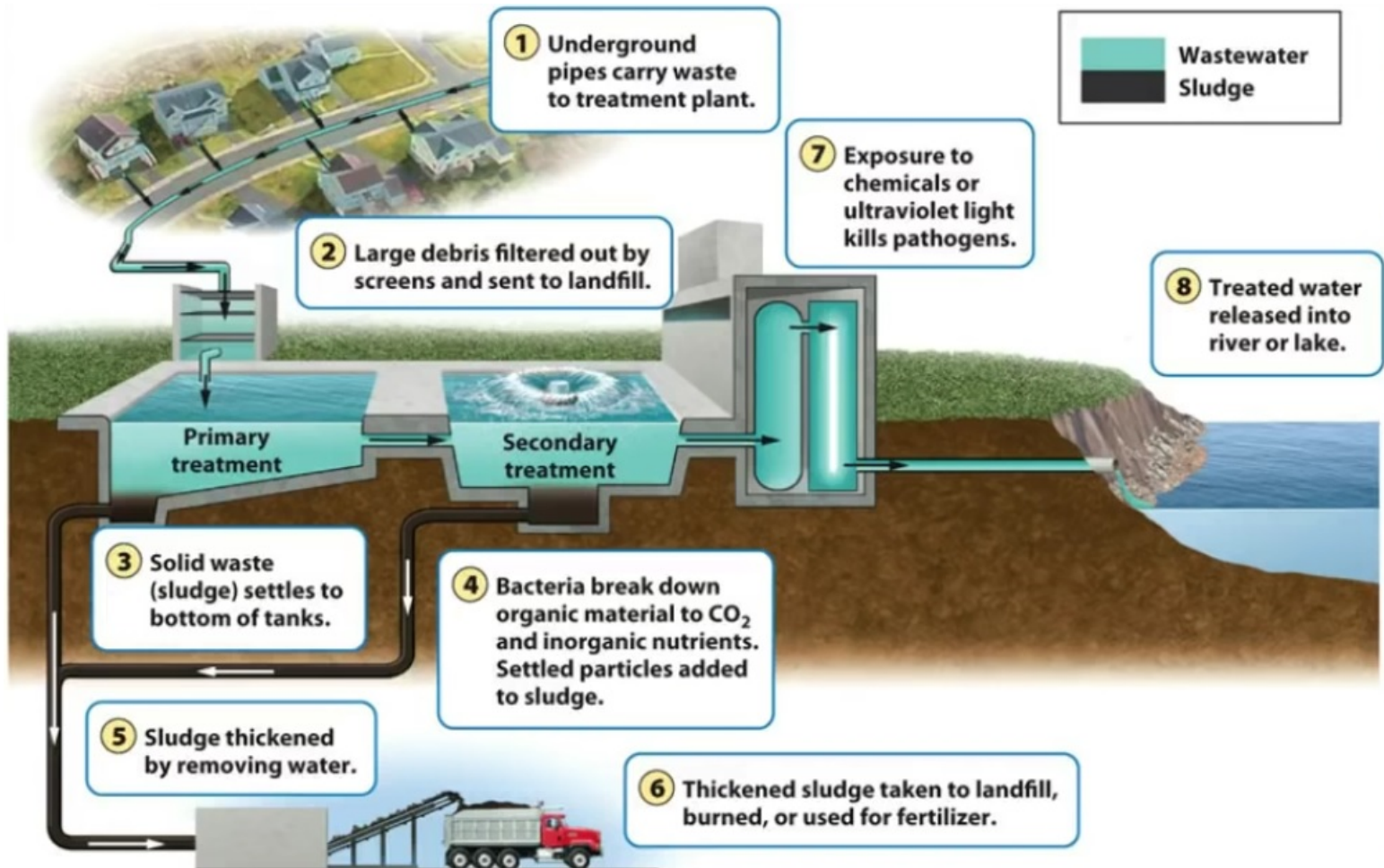
Desalination





Water Resource

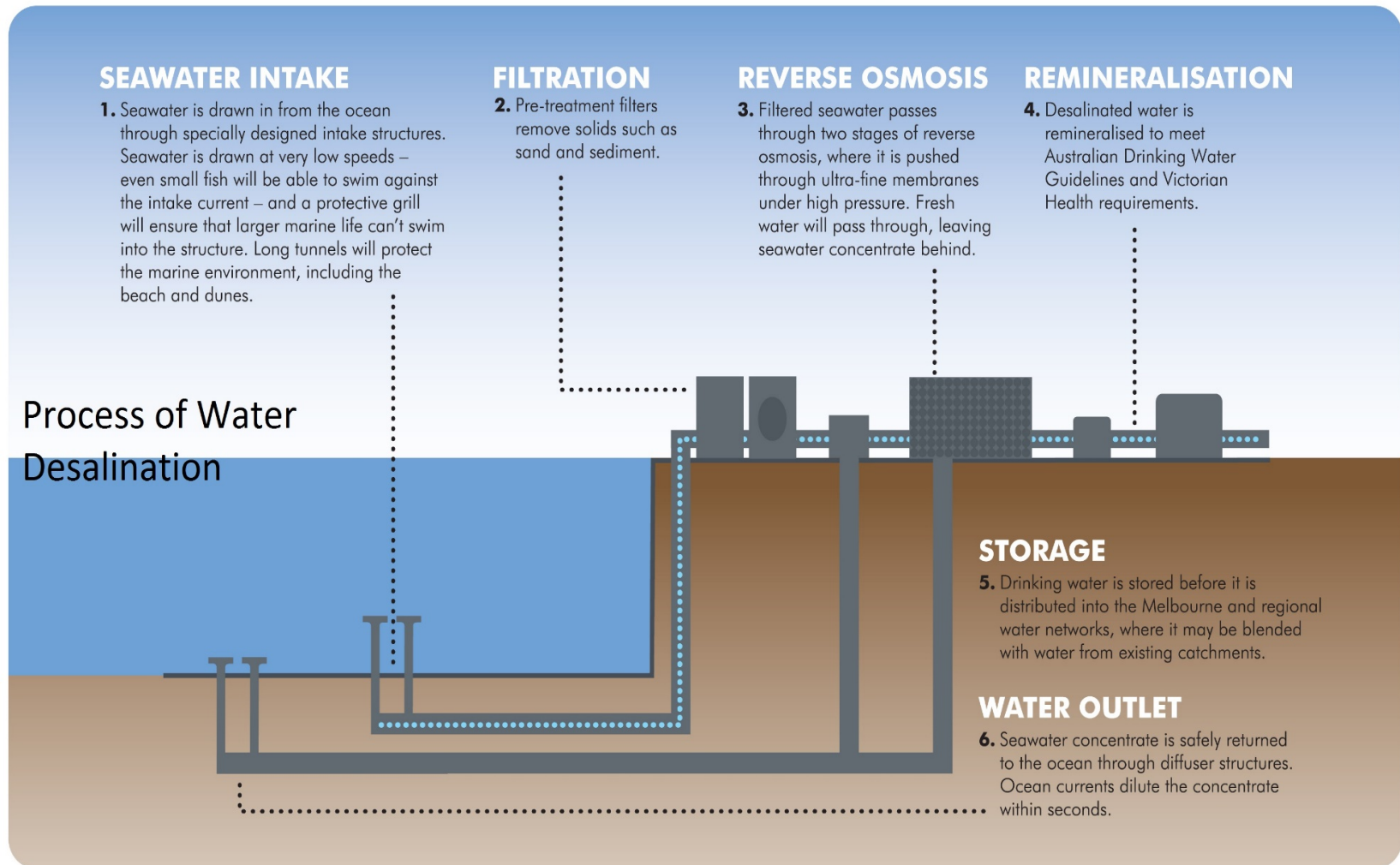
Sewage Treatment Plant





Water Resource

Sewage Treatment Plant





Water Resource

Treatments for Human and Animal Wastewater

- ▣ **Septic systems-** a large container that receives wastewater from the house. Anything that floats to the top is “scum.” Anything that sinks is “sludge.” The middle layer is clear water called “septage.” The septage contains bacteria, but also pathogenic organisms, N and P. The water flows through pipes and lawn called the leach field.

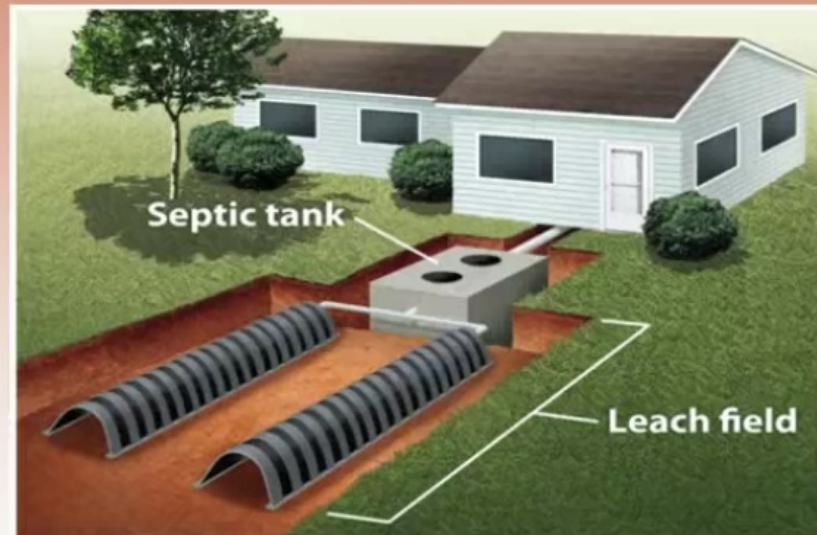


Figure 14.5
Environmental Science
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Pro - No electricity is used;
con - must be pumped out
every 5-10 years



Water Resource

Water Movement Water Conservation

Ground water Recharge – - low cost water through subsidies incentives

Natural

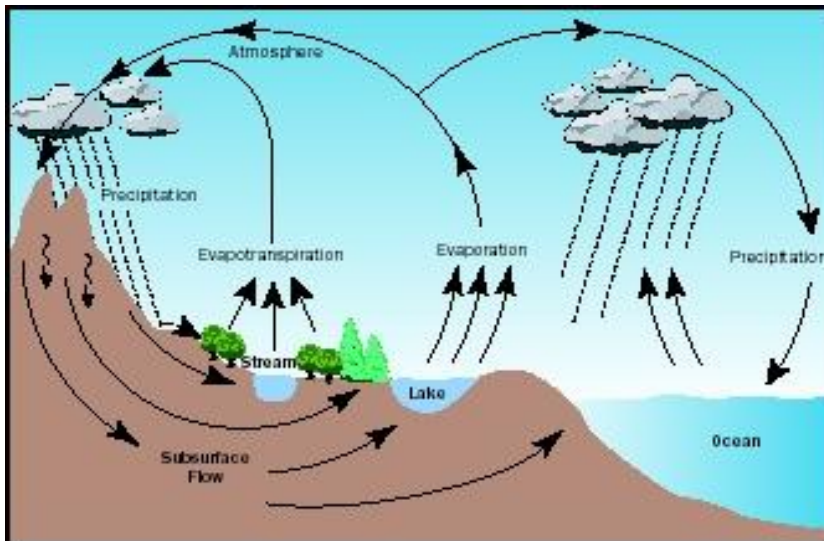
-Water-saving Devices

Saturated flow

Irrigation

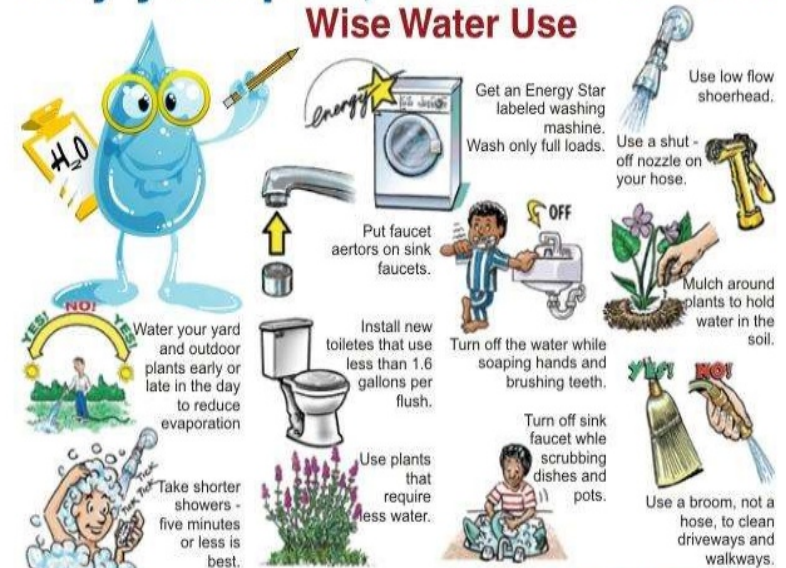
Potable water supply

Desalination



Play your part, be water smart!

Wise Water Use



Save it, or do without it!



**Thank you for your
attention**

Welcoming Q & A !!